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19. The electronic configuration of Einsteinium is : (Given atomic number of Einsteinium = 99)

(1)

(2)

(3)

(4)

20. Match List I with List II

A.		I.	
B.		II.	
C.		III.	
D.		IV.	

Choose the correct answer from the options given below :

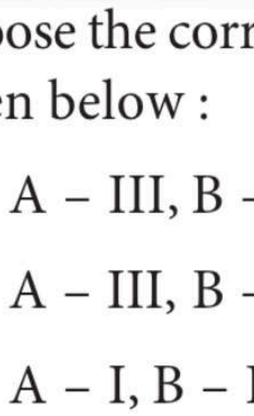
(1) A – III, B – I, C – II, D – IV

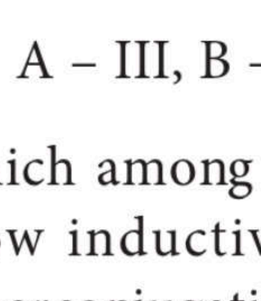
(2) A – III, B – II, C – IV, D – I

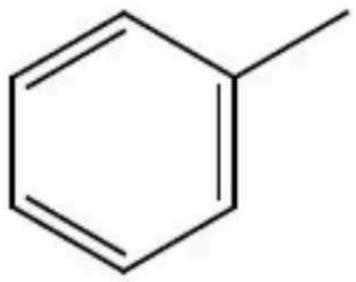
(3) A – I, B – III, C – II, D – IV

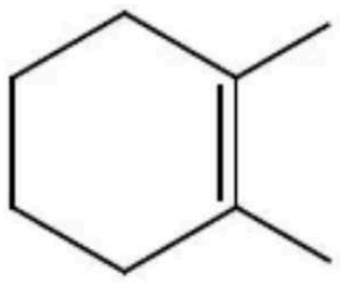
(4) A – III, B – I, C – IV, D – II

21. Which among the following compounds show inductive, mesomeric as well as hyperconjugation effects?

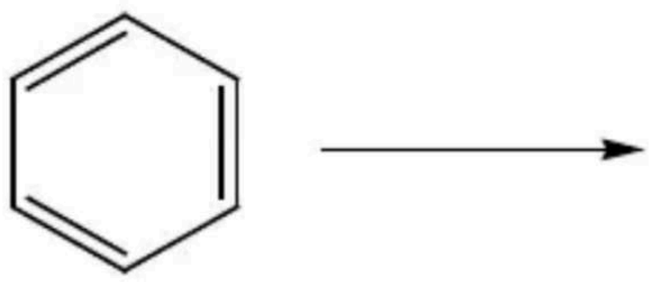
(1) 

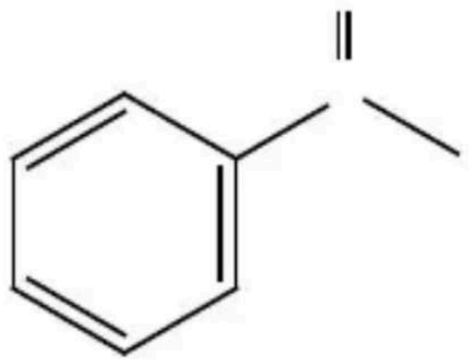
(2) 

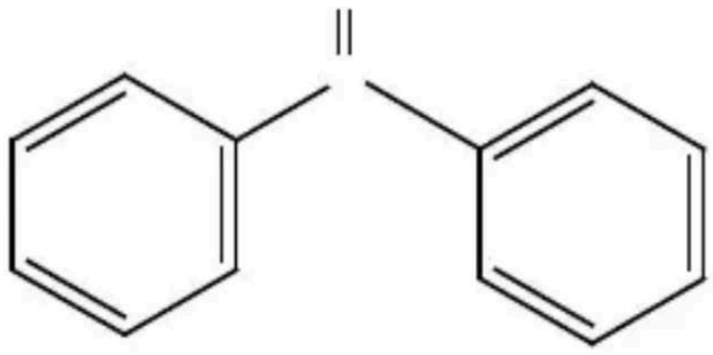
(3) 

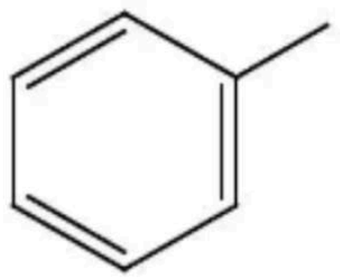
(4) 

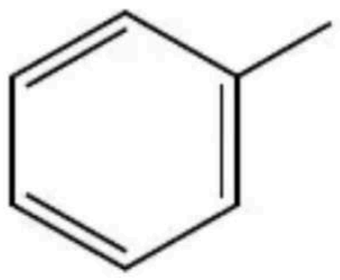
22. Identify major product "X" formed in the following reaction :



(1) 

(2) 

(3) 

(4) 

23. An organic compound containing sulphur gives violet colour in Lassaigne's test. Formula of the violet coloured species formed is

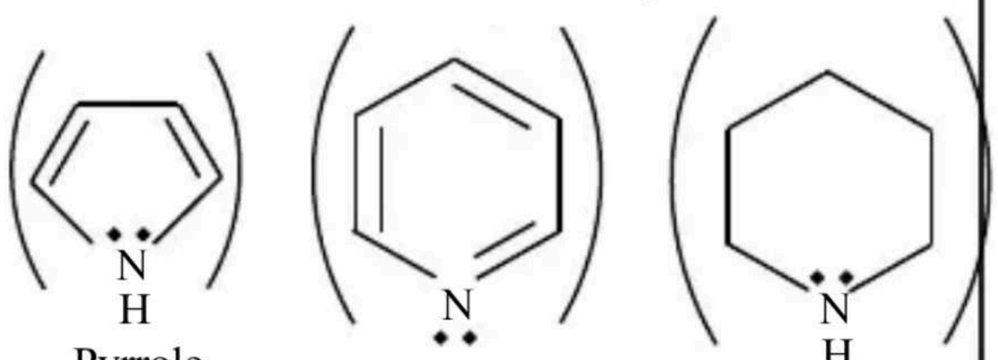
(1)

(2)

(3)

(4) PbS

24. Correct order of basic strength of



Pyrrole Pyridine Piperidine

(1) Piperidine > Pyridine > Pyrrole

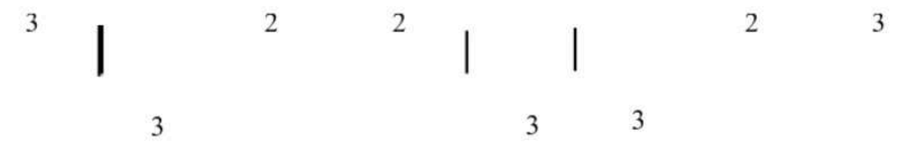
(2) Pyrrole > Pyridine > Piperidine

(3) Pyridine > Piperidine > Pyrrole

(4) Pyrrole > Piperidine > Pyridine

25.

26. IUPAC name of following hydrocarbon is



(1) 2 – Ethyl – 3, 6 - dimethylheptane

(2) 2 – Ethyl –2, 6 - diethylheptane

(3) 2, 5, 6 - Trimethyloctane

(4) 3, 4, 7 - Trimethyloctane

27. The compound which not only gives foul smell when heated with
but also forms a stable diazonium salts is

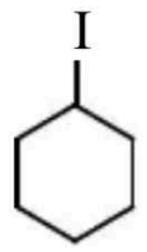
(1) Phenyl methanamine

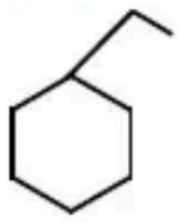
(2) N-methyl methanamine

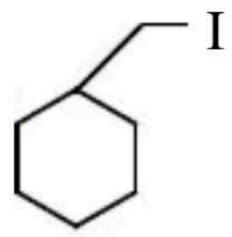
(3) Ethylamine

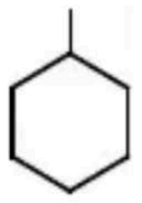
(4) Aniline

28. Which among the following compounds will undergo fastest S_N reaction

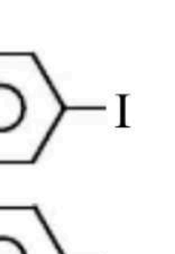
(1) 

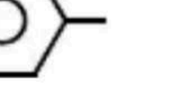
(2) 

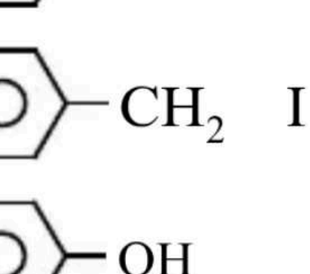
(3) 

(4) 

29. Schiff's base can be represented as:

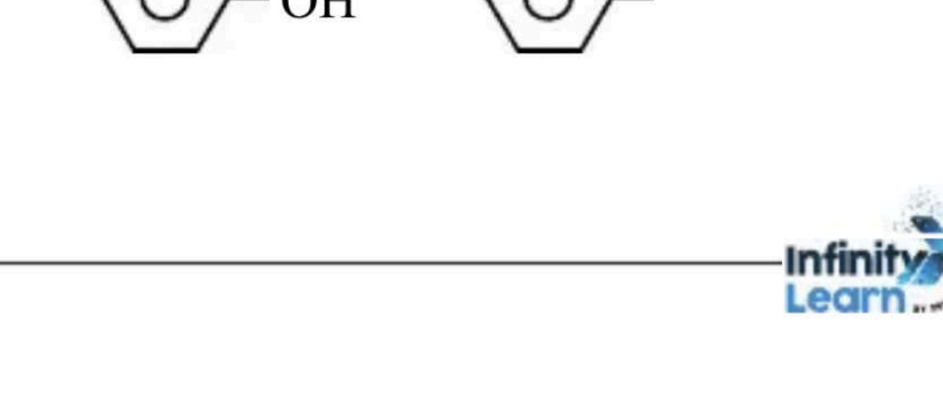
(1) 

(2) 

(3) 

(4)

3



31. Match the oxide given in column A with its property given in column B.

SO

Which of the following options has all correct pairs ?

(1) c c b b
(2) c d b a
(3) c d a b
(4) c a d b

32. For the given reaction, choose the correct expression of K_C from the following :

$Fe_{aq} \quad SCN_{aq} \quad FeSCN \quad aq$

(1) $K_C \quad \frac{Fe\ SCN}{Fe \quad SCN}$

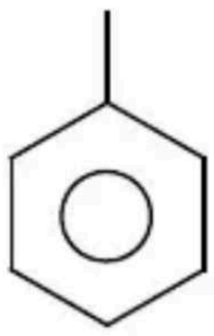
(2) $K_C \quad \frac{Fe \quad SCN}{Fe\ SCN}$

(3) $K_C \quad \frac{Fe\ SCN}{Fe \quad SCN}$

(4) $K_C \quad \frac{Fe\ SCN}{Fe \quad SCN}$

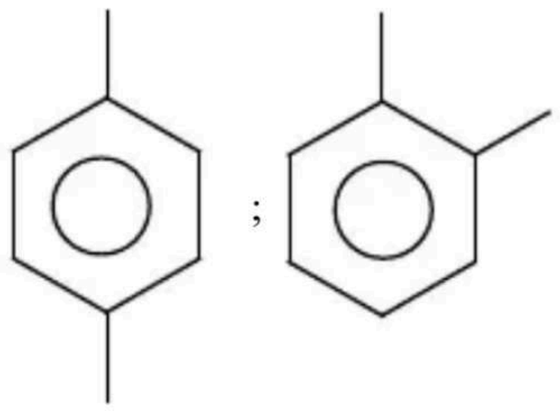
33. Regarding ozone incorrect statement is
(1) It is thermodynamically highly stable
(2) Bond angle is
(3) Bond length is 128 pm
(4) Formal charge on central oxygen atom is +1

34.



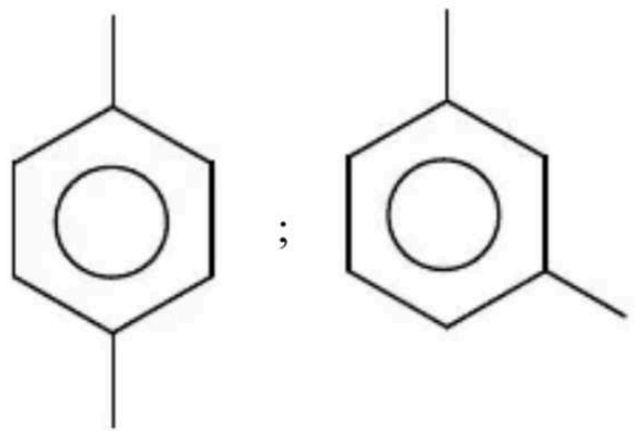
Products A and B, respectively, are

(1)

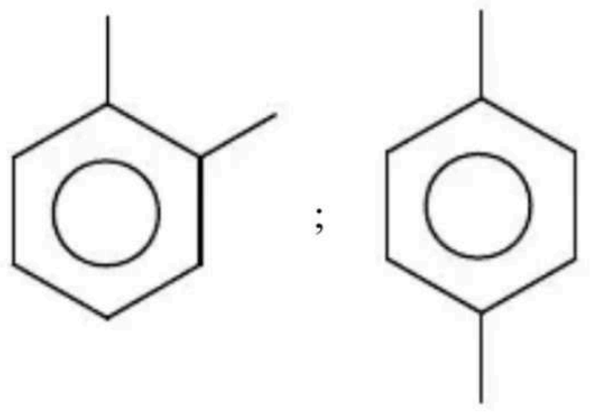


(2)

(3)



(4)



35. Cassiterite is

(1) PbS (2)

(3) (4)

SECTION – B

36. Ratio of $\frac{T_b}{K_b}$ of 10g AB and 14 g $A\ B$ per 100g of solvent in their respective solution (AB and $A\ B$ both are non electrolytes) is 1 mole/kg in both cases. Hence atomic weight of A and B are

(1) 100, 40 (2) 60, 20

(3) 20, 60 (4) 80, 20

37. The correct order in which the O – O bond length increases in the following

(1) $H\ O \quad O \quad O$

(2) $O_3 < H_2O_2 < O_2$

(3) $O_2 < H_2O_2 < O_3$

(4) $O_2 < O_3 < H_2O_2$

38. Which of the following does not contain NH group in its structure?

(1) Proline (2) Glycine

(3) Alanine (4) Tyrosine

39. Neutral l is used in the detection of

(1) Acetone

(2) Ethyl alcohol

(3) Phenol

(4) Acetaldehyde

40. CaF has a solubility of g per 100 mL of solution at 25°C. Its solubility product is (Atomic mass Ca = 40, F = 19)

(1) (2)

(3) (4)

41.

(2) SHE is anode and Cu is cathode

(3) Reduction occurs at SHE electrode

(4) Oxidation occurs at Cu electrode

42. When neutral or faintly alkaline $KMnO_4$ is treated with sodium thiosulphate, thiosulphate ion is converted into :

SO

$SO \qquad S\ O$

43. Given below are two statements:

The higher oxidation states are more stable down the group among transition elements unlike p-block elements.

Interstitial compounds are chemically inert

In the light of the above statements, choose the correct answer from the options given below :

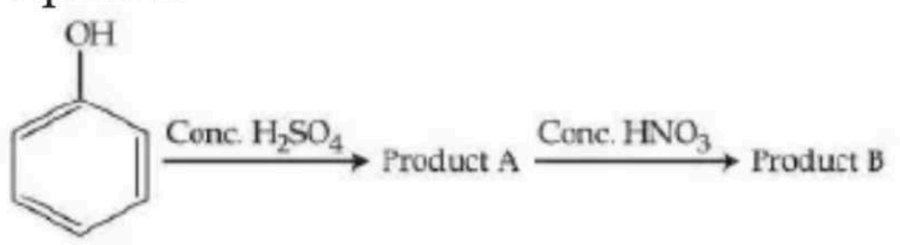
(1) Both statement I and statement II are false

(2) Statement I false but statement II is true

(3) Both statement I and statement II are true

(4) Statement I is true but statement II is false

44. Consider the given chemical reaction sequence



Total sum of oxygen atoms in Product A and Product B are _____

(1) 12

(2) 14

(3) 10

(4) 6

45. Identify the correct statements about p-block elements and their compounds

(A) Non metals have higher electronegativity than metals

(B) Non metals have lower ionization enthalpy than metals

(C) Compounds formed between highly reactive nonmetals and highly reactive metals are generally ionic.

(D) The non-metal oxides are generally basic in nature

(E) The metal oxides are generally acidic or neutral in nature.

(1) (D) and (E) only

(2) (A) and (C) only

(3) (B) and (E) only

(4) (B) and (D) only

46. The complex which contains even number of electrons in orbitals is

(1)

(2)

(3)

(4)

47. The compound which can undergo dehydration with 20% at 358 K is

(1)

(2)

(3)

(4)

48. In the following reactions,

.

The number of sigma bonds present in the product (B) is

(1) 16 (2) 12

(3) 18 (4) 3

49. The incorrect statement regarding chirality is

(1) Enatiomers are superimposable mirror images of each other.

(2) A racemic mixture shows zero optical rotation.

(3) S_N reaction yields 1 : 1 mixture of both enantiomers

(4) The product obtained by S_N reaction of haloalkane having chirality at the reactive site shows inversion of configuration

50. .

'X' is also used in the following conversion

(1)

(2)

(3)

(4)

PHYSICS

Section-A

51. The measured value of the length of a simple pendulum is 20 cm with 2 mm accuracy. The time for 50 oscillations was measured to be 40 seconds with 1 second resolution. From these measurements, the accuracy in the measurement of acceleration due to gravity is

(1) 6% (2) 4%

(3) 8% (4) 5%

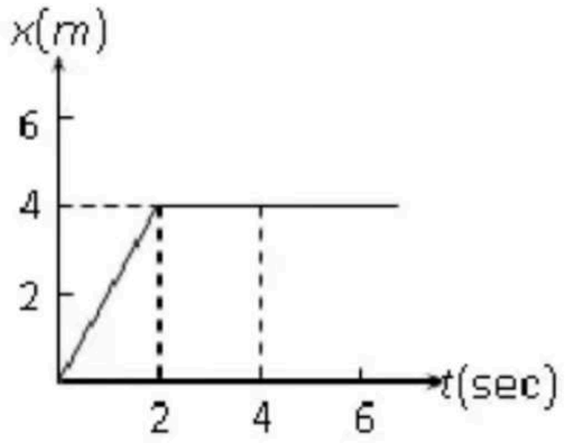
52. From a certain height 'h' two bodies are projected horizontally each with velocity v. One body is projected towards north, and the other body is projected towards east. Their separation on reaching the ground, is

(1) $\sqrt{\quad}$ (2) $\sqrt{\quad}$

(3) $\sqrt{\quad}$ (4) $\sqrt{\quad}$

53. In the figure given below, the position-time graph of a particle of mass 0.1 Kg is shown.

The impulse at t = 2 sec is ( )



(1) 0.2 (2) -0.2

(3) 0.1 (4) -0.4

54. A block of mass m is placed at rest on an inclined plane of inclination to the horizontal. If the coefficient of friction between the block and the plane is , then the total force the inclined plane exerts on the block is

(1) mg (2) *mg*

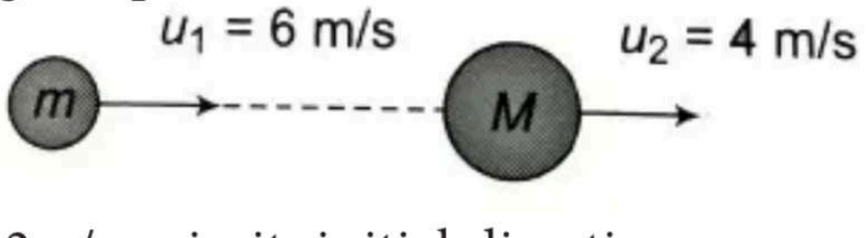
(3) *mg* (4) *mg*

55. If the heart pushes 1 cc of blood in one second under pressure 20000 N/m^2 the power of heart is

(1) 0.02 W (2) 400 W

(3) 5×10^{-10} W (4) 0.2 W

56. A particle of mass m moving with horizontal speed 6 m/s collides another body of mass M as shown in figure. If *m* *M* then for one dimensional elastic collision, the speed of lighter particle after collision will be



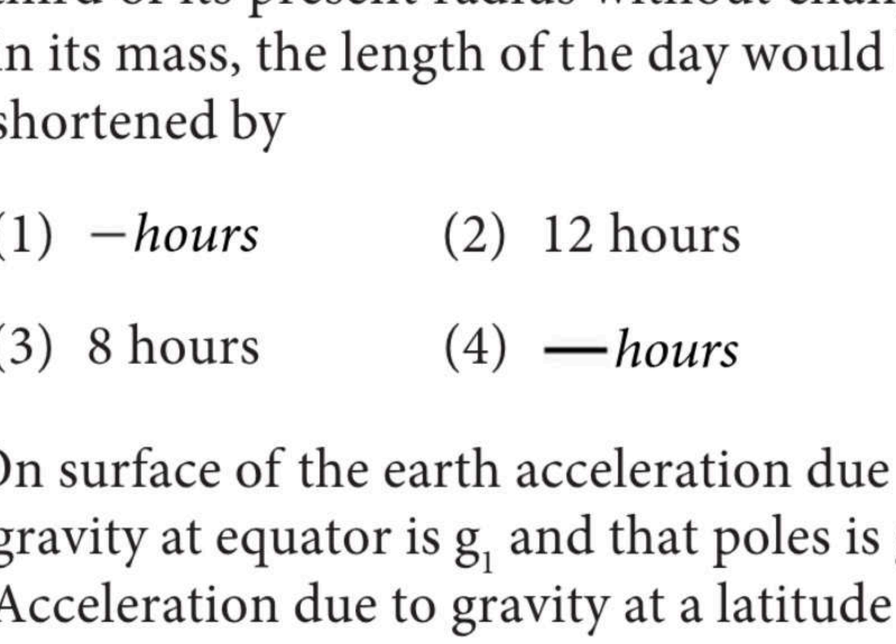
(1) 2 / in its initial direction

(2) 2 / opposite to its initial direction

(3) 4 / opposite to the its initial direction

(4) 4 / in its initial direction

57. Two persons A and B of equal height are carrying a long uniform wooden beam of length l. They are at distance $\frac{l}{-}$ and $\frac{l}{-}$ respectively from the nearest end of the rod as shown. The ratio of normal reaction on the heads of A and B is



(1) 2 : 3 (2) 4 : 9

(3) 5 : 9 (4) 6 : 5

58. If the earth suddenly contracts to one third of its present radius without change in its mass, the length of the day would be shortened by

(1) −*hours* (2) 12 hours

(3) 8 hours (4) —*hours*

59. On surface of the earth acceleration due to gravity at equator is g_1 and that poles is g_2. Acceleration due to gravity at a latitude of 45^0 will be

(1) *g* *g* (2) $\frac{g \quad g}{\quad}$

(3) $\frac{g \quad g}{\quad}$ (4) *g* $\frac{g}{\quad}$

60. A force of 6.4 N stretches a vertical spring by 0.1 m. The mass that must be suspended from the spring so that it oscillates with a period of — sec is

(1) — (2)

(3) — (4)

61. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

If the volume of a body remains unchanged when subjected to tensile strain, the value of Poisson's ratio is 0.5.

The volume of rigid body does not change due to external deformation force.

In the light of the above statement, choose the correct answer from the options below:

(1) (A) is false but (R) is true

(2) (A) is true but (R) is false

(3) Both (A) and (R) are true and (R) is the correct explanation of (A)

(4) Both (A) and (R) are true but (R) is not the correct explanation of (A)

62. The height to which a cylindrical vessel be filled with a homogeneous liquid, to make the average force with which the liquid presses the side of the vessel equal to the force exerted by the liquid on the bottom of the vessel, is equal to

(1) Half of the radius of the vessel

(2) Radius of the vessel

(3) One-fourth of the radius of the vessel

(4) Three-fourth of the radius of the vessel

63. Excess pressure inside a soap bubble is three times that of the other soap bubble, then the ratio of their volumes will be

(1) 1 : 3 (2) 1 : 9

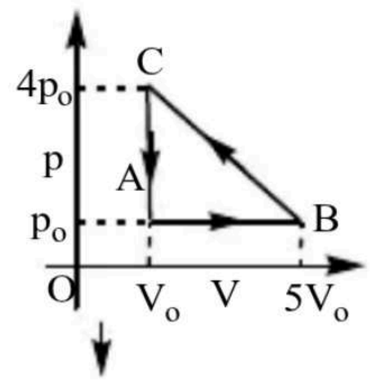
(3) 1 : 27 (4) 1 : 81

64. Certain quantity of water cools from 70°C to 60°C in the first 5 minutes and to 54°C in the next 5 minutes. The temperature of the surrounding is

(1) 45°C (2) 20°C

(3) 42°C (4) 10°C

65. An ideal monatomic gas is taken from state A to B, then from state B to state C and finally back to state A as shown in the P-diagram. Which of the following is correct for BC process?



(1) Q +ve, U -ve, W +ve

(2) Q +ve, U +ve, W –ve

(3) Q +ve, U +ve, W +ve

(4) Q -ve, U -ve, W –ve

66. A cylinder of capacity 20 litres is filled with H gas. The total average kinetic energy of translatory motion of its molecules is J. The pressure of hydrogen in the cylinder is

(1) $N\ m$ (2) $N\ m$

(3) $N\ m$ (4) $N\ m$

67. Two open organ pipes of length 25 cm and 25.5 cm produce 10 beat/sec. The velocity of sound in the gas enclosed in pipes will be

(1) 255 (2) 250

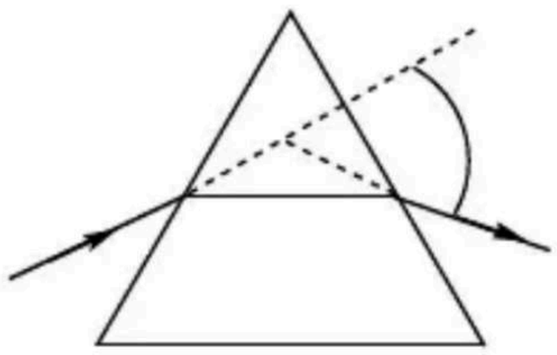
(3) 350 (4) 345m/s

68. In the experiment for the determination of the speed of sound in air using the resonance column method, the length of the air column that resonates in the fundamental mode, with a tuning fork is 0.1 m. When this length is changed to 0.35 m, the same tuning fork resonates with the first overtone. Calculate the end correction.

(1) 0.012 m (2) 0.025 m

(3) 0.05 m (4) 0.024 m

69. The angle of deviation through a prism is minimum when



(A) Incident ray and emergent ray are symmetric to the prism

(B) The refracted ray inside the prism becomes parallel to its base

(C) Angle of incidence is equal to that of the angle of emergence

(D) When angle of emergence is double the angle of incidence

Choose the correct answer from the options given below:

(1) Only statements (A) and (B) are true

(2) Only statement (D) is true

(3) Statements (B) and (C) are true

(4) Statements (A), (B) and (C) are true

70. Consider the diffraction pattern obtained from the sunlight incident on a pinhole of diameter 0.1 μm. If the diameter of the pinhole is slightly increased, it will affect the diffraction pattern

(1) Width of fringes decreases, but intensity increases

(2) Width of fringes decreases, but intensity decreases

(3) Width increases, but intensity decreases

(4) Width increases, but intensity increases

71. In a Young's double slit experiment two slits are separated by 2 mm and the screen is placed one meter away. When a light of wavelength 500 nm is used, the fringe separation will be:

(1) 1 mm

(2) 0.25 mm

(3) 0.75 mm

(4) 0.50 mm

72. Choose the incorrect one/ones from the statements given below

(a) The electric lines of force entering into a Gaussian surface provide negative flux

(b) A charge 'q' is placed at the centre of a cube. The flux through all the faces will be the same

(c) In a uniform electric field net flux through a closed Gaussian surface containing no net charge, is zero

(d) When the electric field is parallel to a Gaussian surface, it provides a finite non-zero flux.

(1) b and only (2) d only

(3) c and d only (4) a and c only

$$r \quad ai \quad bj$$

$$E \quad E\,i \quad E\,j$$

$$Q\sqrt{E\,a \qquad E\,b}$$

$$Q \;\; E \quad E \;\sqrt{a \quad b}$$

$$Q\sqrt{E \quad E} \;\sqrt{a \quad b}$$

74. A steady current flow in a metallic conductor of non-uniform cross-section. The quantity/ quantities constant along the length of the conductor is/are

(1) Current, electric field and drift speed

(2) Drift speed only

(3) Current and drift speed

(4) Current only

75. The resistance per centimeter of a meter bridge wire is r, with resistance in left gap. Balancing length from left end is at 40 cm with resistance in right gap. Now the wire is replaced by another wire of 2r resistance per centimeter. The new balancing length from left for same settings will be

(1) 20 cm (2) 40 cm

(3) 80 cm (4) 10 cm

x
z

—

78. A rectangular loop of length 2.5 m and width 2 m is placed at to a magnetic field of 4 T. The loop is removed from the field in 10 s. The average emf induced in the loop during this time is:

79. What happens to the inductive reactance and the current in a purely inductive circuit if the frequency of AC is halved?

80. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : In a matter wave both position and momentum cannot be measured accurately at the same time.

Reason (R) : If P is uncertainty in momentum and x is uncertainty in position, then according to principle of uncertainty $P\ x\ \frac{h}{}$ (for more rigorous treatment).

In the light of the above statement, choose the correct answer from the options below:

(1) (A) is false but (R) is true

(2) (A) is true but (R) is false

(3) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(4) Both (A) and (R) are true and (R) is the correct explanation of (A)

81. The velocity of the most energetic electrons emitted from a metallic surface is doubled when the frequency ' ' of the incident radiation is doubled. The work function of the metal is

(1) $\frac{h}{}$

(2) $\frac{h}{}$

(3) $\frac{h}{}$

(4) $-h$

82. Which of the following statements are correct?

(4) $1.5 \times 10^5\ \mathrm{Vm}^{-1}$ from p to n side

84. An object is placed in a medium of refractive index 3. An electromagnetic wave of intensity $6\times10^8\,\mathrm{W/m^2}$ falls normally on the object and it is absorbed completely. The radiation pressure on the object would be (speed of light in free space)

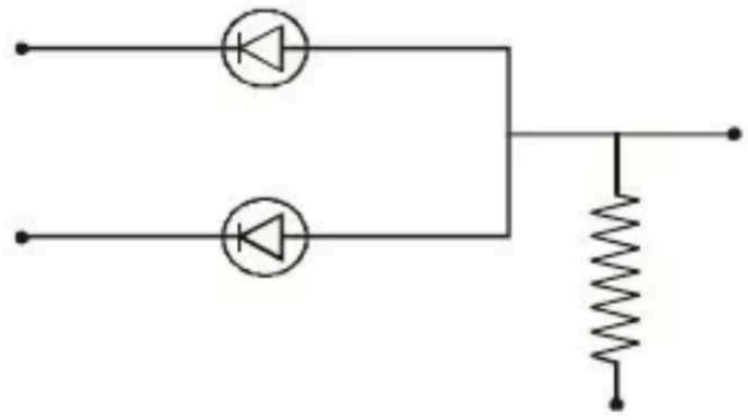
(1) $36\mathrm{Nm}^{-2}$

(2)

(3)

(4) $2\,\mathrm{Nm}^{-2}$

85. In the circuit, for two inputs A and B, output is shown. The circuit will indicate



(1) AND Gate
(2) OR Gate
(3) NOT Gate
(4) NAND Gate

SECTION - B

86. In a Screw Gauge, fifth division of the circular scale coincides with the reference line when the ratchet is closed. There are 50 divisions on the circular scale and the main scale moves by on a complete rotation. For a particular observation the reading on the main scale is and the 20th division of the circular scale coincides with reference line. Calculate the true reading.

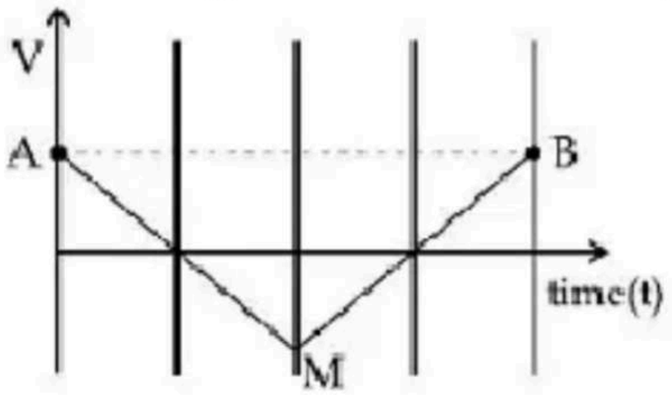
(1)

(2) *mm*

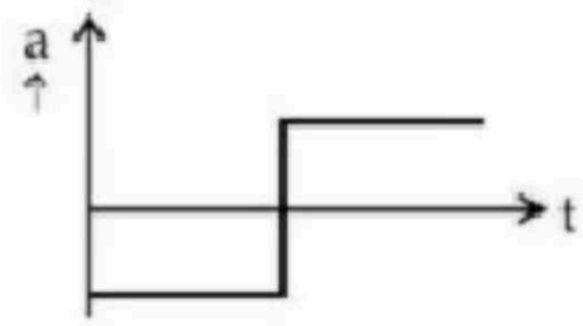
(3) *mm*

(4) *mm*

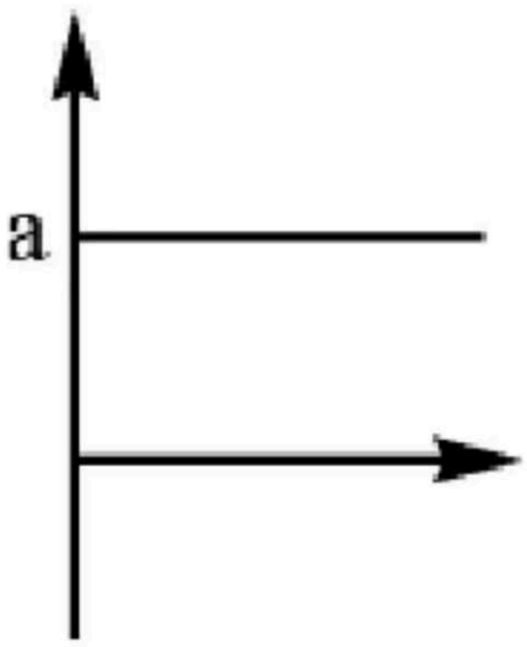
87. If the velocity time graph has the shape AMB, what would be the shape of the corresponding acceleration-time graph?



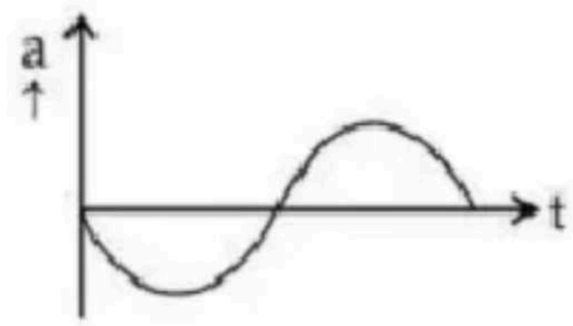
(1)



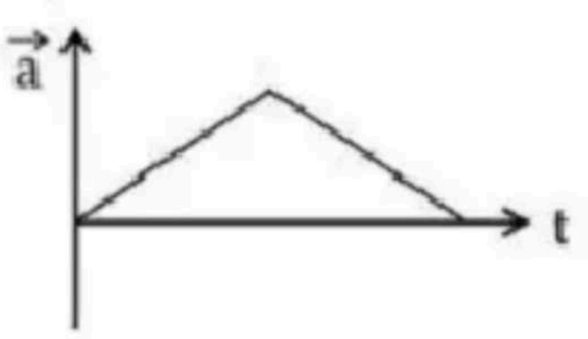
(2)



(3)



(4)



88. A satellite is revolving round the earth with orbital speed . If it stops suddenly, the speed with which it will strike the surface of earth would be : (v_e = escape velocity of a particle on earth's surface)

(1) $\frac{v_e}{v}$ (2) v

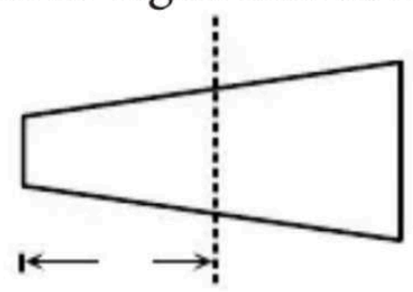
(3) $\sqrt{v_e - v}$ (4) $\sqrt{v_e - v}$

89. If x a t — and $x' = a$ ωt, then what is the phase difference between the two waves?

(1) $\pi / 3$ (2) $\pi / 6$

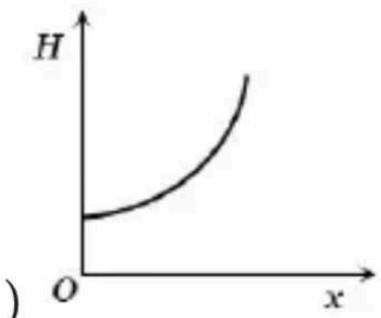
(3) $\pi / 2$ (4) π

90. Radius of a conductor increases uniformly from left end to right end as shown in fig.

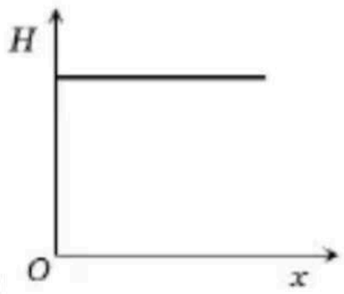


Material of the conductor is isotropic and its curved surface is thermally isolated from surrounding. Its ends are maintained at temperatures $_1$ and $_2$ ($_1$ > $_2$). If, in steady state, heat flow rate is equal to , then which of the following graphs is correct for variation of H with distance 'x' from T_1 end?

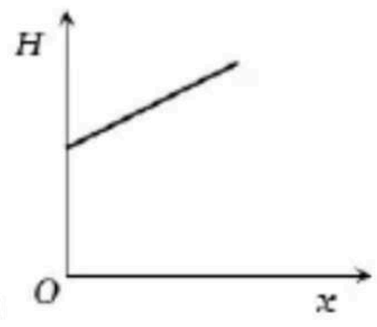
(1)



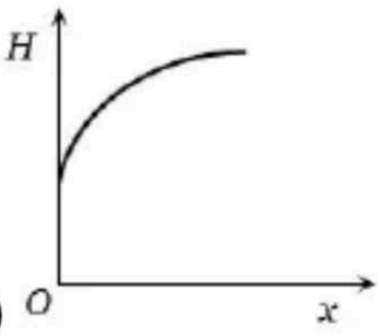
(2)



(3)



(4)



91. When an isolated gas expands against vacuum, then

(a) No work is done in the process

(b) Internal energy does not change in the process

(c) The process is like adiabatic

(1) both a and b are true

(2) both b and c are true

(3) both a and c are true

(4) a, b, c are true

92. Bodies of masses 4 kg and 2 kg are attached at the ends of a massless rod. The positions of 4 kg and 2 kg are $\hat{i}+\hat{j}$ and + Then the position of centre of mass of the system is

(1) $\left(\frac{\hat{i}+\hat{j}}{}\right)$

(2) $\left(\frac{\hat{i}+\hat{j}}{}\right)$

(3) $\frac{i \quad j}{}$

(4) $\left(\frac{\hat{i}-\hat{j}}{}\right)$

93. **Assertion-(A) :** Sound produced by an open organ pipe is richer than the sound produced by a closed organ pipe.

Reason-(R) : Outside air can enter the pipe from both ends, in case of open organ pipe.

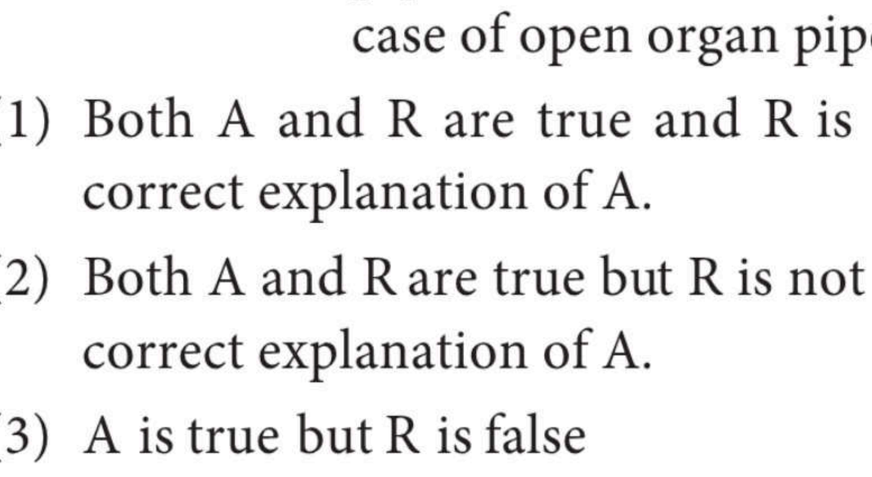
(1) Both A and R are true and R is the correct explanation of A.

(2) Both A and R are true but R is not the correct explanation of A.

(3) A is true but R is false

(4) Both A and B are false.

94. Find the distance of image from object O, formed by the combination of lenses in the figure .



(1) 20 cm

(2) 10 cm

(3) infinity

(4) 75 cm

95. The electrostatic field (E) between the metal plates of an isolated parallel plate capacitor C having a charge Q and area A, is

(1) independent of the distance between the plates

(2) linearly proportional to the distance between the plates

(3) inversely proportional to the distance between the plates

Ω

97. In series LCR resonance circuit, if we change the resistance from a lower to higher value

(1) The quality factor and the resonance frequency will remain constant

(2) The quality factor will increase

(3) The bandwidth of resonance circuit will increase

(4) The resonance frequency will increase

98. The potential energy of a particle of mass is given by

$$E \qquad x$$

$$U\ x \qquad\qquad x$$

$$\lambda \qquad \lambda \qquad\qquad \leq x \leq \qquad x$$

$$E_0 \qquad \frac{\lambda}{\lambda}$$

$$\sqrt{\qquad} \qquad\qquad \sqrt{\qquad}$$

$$(\mu)$$

100. One requires an energy to remove a nucleon from nucleus and an energy to remove an electron from an atom. then

$$\leq$$

BOTANY

Section-A

101. Why is it advantageous for a vector to have a single recognition site for a restriction enzyme when cloning foreign DNA?
   (1) It allows the integration of multiple foreign DNA fragments simultaneously.
   (2) It prevents the loss of any antibiotic resistance genes.
   (3) It simpli es the gene cloning process by avoiding the generation of multiple fragments.
   (4) It enhances the vector's ability to replicate within host cells.

102. How many turns of the Calvin cycle are necessary to synthesize one molecule of glucose?
   (1) 3 turns
   (2) 6 turns
   (3) 1 turns
   (4) 12 turns

103. Read the following statements about the processes in photosynthesis and select how many are correct:
   a) Water molecule splitting on the inner side of the thylakoid membrane results in proton accumulation in the lumen.
   b)
   c)
   d)
   iff
   (1) One statement is correct.
   (2) Two statements are correct.
   (3) Three statements are correct.
   (4) Four statements are correct.

104. The source of phosphate for phosphorylation of PGAL in glycolysis is
   (1) ATP
   (2) GTP
   (3) Inorganic phosphate
   (4) ADP

105. Match the following :

	List-I		List-II
A.	Disarmed plasmid	I.	Enables the bacteria to take up the recombinant DNA
B.	Disarmed retroviruses	II.	Plant cells are bombarded with high velocity micro particles of gold / tungsten
C.	Microinjection method	III.	rDNA is directly introduced into nucleus of animal cell
D.	Gene gun method	IV.	Vector used to transfer desired genes into plant cells

Choose the correct answer from the options given below :

(2) A-III, B-I, C-II, D-IV, E-V

(3) A-IV, B-V, C-III, D-II, E-I

(4) A-III, B-II, C-I, D-IV, E-V

106. How many chromatids are present in a single metaphase chromosome?

(1) One

(2) Two

(3) Four

(4) Five

107. Consider the following statements about genetics in a diploid organism:

Statement-I : A diploid organism can have a maximum of two alleles for the ABO blood group system at a time.

Statement-II : Incomplete dominance is a type of interaction that occurs between different alleles(Rr) of the same gene.

Based on these statements, select the correct option:

(1) Both Statement I and Statement II are true

(2) Statement I is false but Statement II is true

(3) Statement I is true but Statement II is false

(4) Both Statement I and Statement II are false

108. Fatty acids enter the respiratory pathway in the form of

(1) PGAL

(2) Acetyl CoA

(3) Pyruvic acid

(4) Amino acid

109.

	List-I (Compound)		List-II (No.of Carbon atoms)
A.	α -keto glutaric acid	I.	2C
B.	Succinic acid	II.	3C
C.	Pyruvic acid	III.	4C
D.	Acetyl group in Acetyl Co A	IV.	5C

Choose the correct answer from the options given below :

(1) A-II, B-I, C-III, D-IV

(2) A-II, B-IV, C-III, D-I

(3) A-IV, B-III, C-I, D-II

(4) A-IV, B-III, C-II, D-I

110. Which statement correctly describes the dominant phase and features of pteridophytes?
(1) The dominant phase is the gametophyte, which has di erentiated vascular tissues and bears sporangia.
(2) The dominant phase is the sporophyte, which is di erentiated into organs such as roots, stems, and leaves and does not have vascular tissues.
(3) The dominant phase is the sporophyte, characterized by well-di erentiated vascular tissues and organs like roots, stems, and leaves.
(4) The gametophyte is the main plant body and it typically forms large structures called strobili or cones.

111. What is the term used to describe the increase in respiration rate observed during the ripening of fruits?
(1) Respiratory climacteric
(2) Pasteur E ect
(3) Emerson's enhancement effect
(4) Warburg effect

112. At which stage of cell division can a pair of synapsed homologous chromosomes, each with four chromatids, be observed for the first time?
(1) Telophase
(2) Metaphase II
(3) Pachytene
(4) Anaphase

List-I **List-II**

List-I		List-II	
A.	Alkaloids	I.	Concanavalin A
B.	Terpenoids	II.	Lemongrass oil.
C.	Essential oils	III.	Monoterpenes, Diterpenes etc.
D.	Lectins	IV.	Vinblastin, curcumin.
E.	Drugs	V.	Morphine, codeine.

Choose the correct answer from the options given below :
(1) A-V, B-III, C-IV, D-II, E-I
(2) A-V, B-III, C-II, D-I, E-IV
(3) A-V, B-IV, C-III, D-II, E-I
(4) A-V, B-II, C-III, D-IV, E-I

114. How do gibberellins a ect the senescence process in fruits ?
(1) They accelerate senescence, allowing fruits to ripen more quickly.
(2) They delay senescence, enabling fruits to remain on the tree longer and extend the market period.
(3) They have no impact on senescence but increase the sugar content in fruits.
(4) They promote senescence in roots and stems, not fruits.

115. How many cells are present in the mature male gametophyte of angiosperms?
(1) One
(2) Three
(3) Five
(4) Seven

116. Which statement about polygenic inheritance is **incorrect**?
(1) It is a type of quantitative inheritance.
(2) It involves multiple genes.
(3) It is in uenced by environmental factors.
(4) It results in only two distinct phenotypic traits.

117. How many ATP molecules are directly synthesized during the glycolysis pathway from one molecule of glucose?
(1) One
(2) Three
(3) Four
(4) Six

118. What type of genetic material is found in Q bacteriophage?
(1) Protein
(2) Single-stranded DNA (ss DNA)
(3) Single-stranded RNA (ss RNA)
(4) Double-stranded DNA (ds DNA)

119. RNA polymerase responsible for transcription of tRNA in Eukaryotes is
(1) RNA polymerase I
(2) RNA polymerase II
(3) RNA polymerase III
(4) RNA polymerase IV

120. *Bt* toxin gene which controls corn borer is
(1) cry I *Ab*
(2) cry II *Ab*
(3) cry I *Ac*
(4) cry II *Ac*

121. During which phase of the cell cycle does major reorganization of cell components take place?
(1) *G* phase
(2) *G* phase
(3) *G* phase
(4) M phase

122. Which components are absent in the stroma lamellae?
(1) Photosystem I (PS I)
(2) Photosystem II (PS II)
(3) NADP reductase
(4) Both Photosystem II (PS II) and NADP reductase

123. In which of the following organisms is extensive compartmentalization of cells absent?
(1) *Chlorella*, a eukaryote
(2) *Chlamydomonas*, a eukaryote
(3) *Nostoc*, a prokaryote
(4) *Paramoecium*, a eukaryote

124. All tissues on the inner side of the endodermis such as pericycle, vascular bundles and pith constitute the
(1) bark
(2) stele
(3) conjunctive tissue
(4) ground tissue

125. Consider the following statements about cellular structures:

Statement-I : The quasi fluid nature of the cell membrane in eukaryotic cells is attributed to the presence of lipid molecules.

Statement-II: Prokaryotic cells lack membrane bound organelles within their cytoplasm and do not have ribosomes.

Based on these statements, select the correct option:
(1) Both Statement I and Statement II are true
(2) Statement I is false but Statement II is true
(3) Statement I is true but Statement II is false
(4) Both Statement I and Statement II are false

126. Pressure gradient in sieve tubes of angiosperms is maintained by
(1) Xylem parenchyma cells
(2) Albuminous cells
(3) Companion cells
(4) Bast fibres

127. What function does the modi ed petiole of the Australian Acacia primarily perform?
(1) Protection
(2) Storage
(3) Respiration
(4) Photosynthesis

128. Evaluate the following statements and choose the correct option:

Assertion A : Angiosperms produce two types of spores and are therefore heterosporous.

Reason R : The typical embryo sac in angiosperms consists of 7 nuclei and 8 cells.

Choose the most appropriate answer based on the statements:

129. Lion, tiger, leopard, housefly, cat and dog belong to same taxonomic category called
(1) Order
(2) Class
(3) Genus
(4) Kingdom

130. Fungus that is extensively used in biochemical genetic work is
(1) *Penicillium*
(2) *Neurospora*
(3) *Agaricus*
(4) *Colletotrichum*

131. Heterosporous vascular pteridophyte that does not produce both fruits and seeds is
(1) *Pinus*
(2) *Cycas*
(3) *Musa*
(4) *Selaginella*

132. Review the following statements and select the appropriate response:

Assertion A : The four daughter cells produced after meiosis are not identical to each other.

Reason R : Crossing over occurs during the pachytene stage of prophase I in meiosis-I.

Determine the correctness of the statements and the relationship between them:
(1) Both A and R are correct and R is the correct explanation of A
(2) Both A and R are correct but R is NOT the correct explanation of A
(3) A is correct but R is not correct

Statement-I: Not all CO_2 molecules produced during the oxidation of glucose in aerobic respiration are released in the mitochondrial matrix.

Statement-II : Cytochromes function as electron carriers within ETS.

Assess the validity of these statements and select the appropriate answer:
(1) Both Statement I and Statement II are true
(2) Statement I is false but Statement II is true
(3) Statement I is true but Statement II is false
(4) Both Statement I and Statement II are false

134. Which of the following is true regarding the formation of false fruits?
(1) Only the ovary contributes to the fruit formation.
(2) The thalamus contributes to the fruit formation
(3) All oral parts including petals and sepals contribute to fruit formation.
(4) False fruits are mainly found in tropical plants like mango and papaya.

	List-I		List-II
A.	Xanthophylls	I.	Blue green
B.	Carotenoids	II.	Yellow green
C.	Chlorophyll *b*	III.	Yellow
D.	Chlorophyll *a*	IV.	Yellow orange

Choose the correct answer from the options given below :

(1) A-IV, B-III, C-I, D-II

(2) A-III, B-II, C-I, D-IV

(3) A-III, B-IV, C-I, D-II

(4) A-III, B-IV, C-II, D-I

SECTION-B

136. Genotype of 'B' blood group is

(a) $^{B}\ ^{B}$ (b) ^{B}

(c) (d) $i^{a}i^{b}$

(1) a or c only (2) b or c only

(3) a or d only (4) a or b only

137. Evaluate the following statements about biological kingdoms and select the correct answer:

Statement-I : Kingdom Protista includes organisms that are autotrophs, saprophytes, and parasites.

Statement-II : Kingdom Fungi exclusively comprises heterotrophic organisms.

In light of the above statements, choose the correct answer from the options given below :

(1) Both Statement I and Statement II are true

(2) Statement I is false but Statement II is true

(3) Statement I is true but Statement II is false

(4) Both Statement I and Statement II are false

138.

	List-I		List-II
A.	Gaseous phytohormone	I.	*IAA*
B.	Carotenoid derivative	II.	*GA*
C.	Adenine derivative	III.	*ABA*
D.	Indole compounds	IV.	*Ethylene*
		V.	*Cytokinin*

Choose the correct answer from the options given below :

(1) A-II, B-I, C-III, D-IV

(2) A-IV, B-III, C-I, D-V

(3) A-IV, B-III, C-V, D-I

(4) A-II, B-I, C-IV, D-III

139. Identify the incorrect match.

(1) bacteriophage–5386 base pairs

(2) *E.coli* - base pairs

(3) Bacteriophage lambda–48502 base pairs

(4) Haploid human DNA– × base pairs

140. Match the following :

	List-I		List-II
(A)	Lactic acid	I)	*Lactobacillus*
(B)	Acetic acid	II)	*Acetobacter aceti*
(C)	Citric acid	III)	*Aspergillus niger*
(D)	Butyric acid	IV)	*Clostridium acetobutylium*
		V)	*Agrobacterium tumefaciens*

(1) A-I, B-II, C-III, D-IV

(2) A-I, B-III, C-II, D-V

(3) A-II, B-I, C-III, D-IV

(4) A-III, B-II, C-I, D-IV

141. **Assertion (A) :** The first stable product formed during CO_2 fixation in the Calvin cycle is a three-carbon compound.

Reason (R) : The compound identified as the first CO_2 fixation product by Melvin Calvin using radioactive carbon was 3-phosphoglyceric acid (PGA), which contains three carbon atoms.

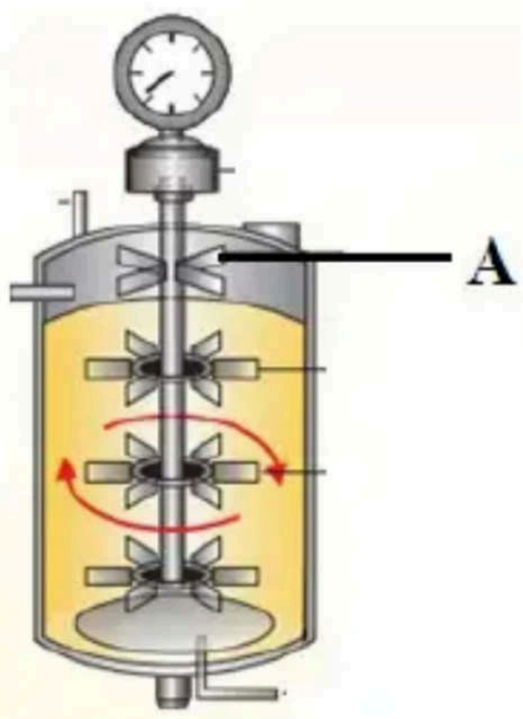
Select the most appropriate answer from the options below:

(1) Both A and R are correct and R is the correct explanation of A

(2) Both A and R are correct but R is NOT the correct explanation of A

(3) A is correct but R is not correct

142. Identify 'A' in the given diagram.



(1) Inlet for raw material

(2) Motor

(3) Flat blade impeller

(4) Foam braker

143. In which types of plants is emasculation not required during hybridization?

(1) Dioecious plants

(2) Monoecious plants

(3) Plants with bisexual flowers

(4) Both dioecious and monoecious plants

144. Which of the following is **incorrect** regarding flower of *Hibiscus*

(1) Anthers are dithecous, and tetrasporangiate

(2) Ovary is superior

(3) Epicalyx is present

(4) Corolla shows twisted aestivation

	List-I	List-II
A.	Dihybrid F_2 phenotypic ratio	I. 1 : 1 : 1 : 1
B.	Monohybrid F_2 phenotypic ratio	II. 9 : 3 : 3 : 1
C.	Incomplete dominance F_2 genotypic ratio	III. 1 : 2 : 1
D.	Dihybrid test cross ratio	IV. 3 : 1

Choose the correct answer from the options given below :

(1) A-II, B-IV, C-III, D-I

(2) A-II, B-III, C-IV, D-I

(3) A-II, B-IV, C-I, D-III

(4) A-II, B-I, C-IV, D-III

146. Why are restriction enzymes important in genetic engineering?

(1) They create sticky ends that facilitate the removal of DNA sequences.

ff